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Abstract

Regularized Adjusted Plus-Minus (RAPM) is a standard approach for estimating player impact in basketball, but it assumes that player contributions are additive and does not account for interactions between players. In practice, team performance often depends on how players perform together, making it important to identify and quantify these interaction effects. However, explicitly modeling interactions within a RAPM framework leads to a combinatorial explosion, making traditional approaches impractical.

In this work, we propose a framework for detecting player interactions using Bayesian Additive Regression Trees (BART). BART models the response as a sum of regression trees and naturally captures interaction effects through hierarchical splitting, while a prior on tree depth limits the complexity of interactions considered. Rather than enumerating all possible interaction terms, we analyze the structure of the fitted trees to identify interactions.

We introduce metrics based on marginal and conditional inclusion proportions to measure how frequently players appear in the model individually and in combination. Using simulated lineup data with a known interaction structure, we demonstrate that this approach can successfully identify meaningful player interactions. We also examine the sensitivity of the method to modeling choices and randomness in the BART fitting procedure.

Keywords: five or six important terms not in the title (and usually not in the abstract either?), multiclass, classification, data augmentation

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1 Introduction

Regularized Adjusted Plus-Minus (RAPM) is a widely used approach in basketball analytics for estimating individual player impact. These models aim to quantify how much a player contributes to their team’s success by analyzing changes in scoring margin across different lineup combinations. While RAPM provides valuable insights, it relies on the assumption that player contributions are additive. In reality, basketball is a highly interactive sport, where the effectiveness of a player often depends on who they are playing with. Some player combinations exhibit strong synergy, while others may underperform despite strong individual talent.

RAPM addresses high dimensionality through regularization but does not explicitly model interactions between players. Extending the model to include interaction terms is theoretically possible; however, the number of potential interactions grows rapidly with the number of players. Even when restricting attention to pairwise interactions, the number of terms increases combinatorially, making estimation computationally difficult and prone to overfitting. As a result, most implementations focus solely on main effects.

More flexible modeling approaches, such as tree-based methods, provide an alternative framework for capturing interactions without explicitly specifying them. In particular, Bayesian Additive Regression Trees (BART) have been introduced as a nonparametric regression method capable of modeling interactions in high-dimensional data. BART represents the response as a sum of regression trees, allowing it to naturally capture interaction effects through hierarchical splitting.

In this work, we propose using BART as a tool to identify and quantify player interactions within a RAPM-style framework. Rather than explicitly including all possible interaction terms, we leverage the structure of BART to implicitly model interactions while controlling their complexity through a prior on tree depth. This prior favors shallow trees, which correspond to lower-order interactions, allowing the model to focus on the most meaningful player combinations.

To detect interactions, we analyze the frequency with which players appear in the BART tree structures. We introduce metrics based on marginal and conditional inclusion proportions to measure how often players are used individually and in combination across trees. By comparing observed co-occurrence rates to what would be expected under independence, we are able to identify pairs of players that exhibit strong interaction effects.

We evaluate this approach using simulated lineup data with known interaction structure. This allows us

to assess whether the proposed methodology can successfully recover true interactions and to explore its sensitivity to modeling choices and randomness in the BART fitting procedure.

2 Methodology

2.1 BART Model

Bayesian Additive Regression Trees (BART) model the response as a sum of regression trees. Specifically, the model takes the form

$$y = \sum_{t=1}^m g_t(X) + \epsilon, \quad \epsilon \sim \mathcal{N}(0, \sigma^2),$$

where each g_t represents a decision tree that partitions the predictor space and assigns a constant value to each terminal node. The ensemble of trees allows BART to capture complex nonlinear relationships and interactions between variables without requiring them to be explicitly specified in advance.

A key feature of BART is the prior placed on tree structure. Let d denote the depth of a node within a tree. The probability that a node splits is given by

$$\mathbb{P}(\text{split at depth } d) = \alpha(1 + d)^{-\beta},$$

where $\alpha \in (0, 1)$ and $\beta > 0$ are hyperparameters controlling tree growth. This prior strongly favors shallow trees, making deeper splits increasingly unlikely.

This prior plays an important role in controlling interaction complexity. Each split in a tree corresponds to conditioning on an additional variable, meaning that deeper trees represent higher-order interactions. For example, a tree of depth two can capture pairwise interactions, while deeper trees are required to represent higher-order combinations. By penalizing tree depth, BART implicitly regularizes the order of interactions, allowing the model to focus on simpler and more interpretable relationships.

2.2 Tree Structure Extraction

To study interactions, we need to understand how variables are used within the fitted BART model. In this work, we use the `dbarts` package to fit the model and access the trees that are generated. Each tree is made

up of a sequence of splits based on different variables, and `dbarts` allows us to see which variables are used in those splits. By looking across many trees, we can track how often each player appears and which players tend to appear together.

This approach is motivated by recent work showing that BART can be used not only for prediction, but also for identifying interactions by examining how variables co-occur within tree structures. In particular, variables that frequently appear together in the same tree provide evidence of a joint effect on the outcome (Marvald and Love, 2024).

We also use the `stochtree` package to better understand how the data moves through the trees. This package provides information about how each observation is grouped into terminal nodes (leaves) across the trees. While we do not directly use these groupings to define interactions, they help confirm how the model is partitioning the data and provide additional context for interpreting the tree structure.

Together, these tools allow us to move beyond treating BART as a black box. Instead, we are able to examine the model’s structure and use it to identify patterns in how players are selected, which forms the basis for our interaction detection approach.

2.3 Interaction Detection

To identify interactions between players, we examine how often variables appear together within the BART tree structures. This approach is motivated by the idea that when two variables frequently appear in the same tree, the model is using their joint effect to explain variation in the response (Marvald and Love, 2024).

We begin by defining the marginal inclusion proportion (MIP) for player i as the proportion of trees in which that player is used in a split. This provides a measure of how important a player is individually within the model. However, MIP alone does not capture interactions.

To detect interactions, we define the conditional inclusion proportion (CIP). For a pair of players i and j , the CIP measures how often player i is used in a tree given that player j is also used. In other words, it quantifies how frequently two players appear together in the same tree. A large value of both $CIP_{i|j}$ and $CIP_{j|i}$ suggests that the pair is important jointly, rather than only individually.

To further distinguish true interactions from strong individual effects, we compare conditional inclusion to marginal inclusion. If two players interact, we expect them to appear together more often than would be explained by their individual importance alone. We capture this by examining the difference between CIP

and MIP. Large positive differences indicate that the joint appearance of the players is more frequent than expected, providing additional evidence of an interaction.

Together, these measures allow us to identify player pairs that are consistently used together by the BART model, forming the basis for detecting meaningful interactions.

2.4 Thresholding

While the marginal and conditional inclusion proportions provide a way to measure variable importance and co-occurrence, a threshold is needed to determine when an interaction is considered meaningful. In this work, we focus on thresholding based on the difference between conditional and marginal inclusion proportions.

For a pair of players i and j , we consider the quantity $CIP_{i|j} - MIP_i$ (and similarly for $CIP_{j|i} - MIP_j$). This difference captures how much more frequently a player appears in a tree when paired with another player than would be expected based on its marginal usage alone.

In practice, we examine thresholds on this difference to identify candidate interactions. Through experimentation on simulated data, we observe that requiring the CIP–MIP difference to exceed a fixed threshold can help distinguish interacting pairs from those that co-occur due to strong individual effects. In particular, a threshold of 0.1 is found to consistently highlight the known interacting pair in many simulation runs.

We note that the choice of threshold is not fixed and can influence the number and stability of detected interactions. Lower thresholds tend to include more candidate pairs, while higher thresholds result in more conservative selection. As a result, thresholding should be viewed as a tuning component of the method rather than a definitive decision rule.

3 Data

[question]: I could give a very brief explanation of the data, but to be honest I never worked with the data, so I feel that I wouldn't be able to give the most adequate description here

4 Data Simulation

4.1 Simulation Setup

To evaluate our proposed approach, we construct a simulated dataset designed to resemble basketball lineup data. Each observation represents a stint, defined as a period of play during which the set of players on the court remains fixed. The response variable represents the point differential during the stint.

We consider a total of $P = 30$ players, consisting of 15 home players and 15 away players. For each stint, five players are randomly selected from each team to be on the court. These lineups are encoded in a design matrix $X \in \mathbb{R}^{S \times P}$, where S is the number of stints. Each row of X corresponds to a stint, with entries equal to +1 for home players on the court, -1 for away players on the court, and 0 otherwise.

The response variable is generated according to a linear model with an additional interaction term. Specifically, we define

$$y = X\beta + \gamma \cdot \mathbb{I}(\text{players 1 and 2 are both on the court}) + \epsilon,$$

where β represents individual player effects, γ controls the strength of the interaction between players 1 and 2, and $\epsilon \sim \mathcal{N}(0, \sigma^2)$ represents noise.

This setup allows us to embed a known interaction between players 1 and 2 while keeping the rest of the model additive. Because the true interaction structure is known, this simulation provides a controlled environment for evaluating whether the proposed method can successfully detect meaningful interactions.

5 Results

We evaluate the proposed interaction detection approach using the simulated dataset described in the previous section. Because the data-generating process includes a known interaction between players 1 and 2, we examine whether the BART-based metrics are able to identify this pair as exhibiting an interaction.

Across runs, players 1 and 2 tend to have relatively high marginal inclusion proportions compared to other players, indicating that they are frequently used in the BART trees. More notably, the conditional inclusion proportions for this pair are often elevated, suggesting that these players appear together in trees more frequently than many other pairs. This pattern is consistent with the presence of a joint effect in the data-generating process.

When comparing conditional inclusion to marginal inclusion, we often observe that the difference between these quantities is larger for players 1 and 2 than for most other pairs. In particular, when using a threshold based on the difference between CIP and MIP, we find that requiring this difference to exceed 0.1 results in players 1 and 2 being identified as an interacting pair across many different random seeds. This suggests that the CIP–MIP difference may be a useful metric for highlighting interactions that are not explained by marginal importance alone.

We also observe that the identification of interactions can be sensitive to the choice of threshold used to define “significant” co-occurrence. In some cases, using a stricter threshold results in fewer detected interactions, and the pair of players 1 and 2 may not always be selected. In contrast, slightly more permissive thresholds tend to include this pair more consistently, along with additional candidate interactions.

Additionally, there is variability across different random seeds. While players 1 and 2 frequently appear among the strongest candidates for interaction, other player pairs may also appear intermittently. This suggests that the stochastic nature of the BART fitting procedure can influence which interactions are identified in a given run.

Overall, these results indicate that the proposed approach is capable of highlighting player pairs that exhibit patterns consistent with interaction effects in the simulated data. At the same time, the variability observed across runs and threshold choices suggests that interaction detection should be interpreted with some caution and may benefit from repeated runs or aggregation strategies.

6 Discussion

References

Marvald, J. and Love, T. (2024). Detecting interactions using bayesian additive regression trees. *Pattern Analysis and Applications*, 27:153.